

Main and Recent Activities

Current Position and Roles

CURRENT POSITION

University of Udine

Udine, Italy

ASSOCIATE PROFESSOR AT AViRES/MLP LABS (UNIVERSITY OF UDINE) - MACHINE LEARNING AND COMPUTER VISION

Dec 2021 - To Date

Recent Activities/Achievements:

- Demonstrated strong academic leadership through roles such as member of the PhD board in Computer Science and AI (since 2021), deputy coordinator of the PhD course in Computer Science and AI (2021-2024), member of the CCS, CAQ, and internship committee for SCMTI (since 2018), while actively teaching various Bachelor, Master, and PhD courses.
- Led and conducted cutting-edge research in computer vision, with a focus on developing novel algorithms for object tracking, target re-identification, image colorization, food analysis, medical imaging, and super resolution.
- Published extensively in top-tier conferences and journals, including ECCV, CVPR, and WACV.
- Actively contributed to the computer vision community by organizing workshops, serving as a program committee member and area chair, and reviewing papers for leading conferences.
- Principal Investigator for the EU Grant 024 2nd Call GREEN ERA-HUB entitled "FarmScan: Field Analysis via Remote Monitoring and Sustainable, Decomposable Agricultural Nodes" (FarmScan)
- Co-Principal Investigator for the ONR Grant N62909-20-1-2075 entitled "Target Re-Association for Autonomous Agents" (TRAAA)
- Applied computer vision techniques to diverse industry domains, including healthcare, retail, food industry, and security, delivering impactful solutions that address real-world challenges.

INTERNATIONAL ROLES

International Conferences, Workshops and Summer Schools in Computer Vision and Machine Learning

Global

GENERAL CHAIR/MEMBER OF THE ORGANIZATION COMMITTEE

Italian IAPR chapter

Firenze, Italy

DEPUTY CHIEF FOR THE NEURAL NETWORKS AND DEEP LEARNING TECHNICAL COMMITTEE

September 2017 - To Date

Creating a network of people that will guide new and young members through the challenges and existing approaches of deep neural networks (deep learning).

NATO

Bruxelles, Belgium

MEMBER OF THE NATO ACTIVITIES IST-169 ROBUSTNESS AND ACCOUNTABILITY IN MACHINE LEARNING SYSTEM &

IST-ET-113 UNSUPERVISED MACHINE LEARNING IN THE MILITARY DOMAIN, & IST-221 DEFENSE AGAINST ADVERSARIAL

2018 - To Date

ATTACKS ON MACHINE LEARNING SYSTEMS SPACE

Determine the state-of-the-art in robustness and accountability of deep learning systems. The work produced out of IST-169 received the Best Paper Award at the IST-190 Research Symposium on AI, ML and BD for Hybrid Military Operations (AI4HMO).

Teaching and Research

TEACHING AT INTERNATIONAL SCHOOLS

IAPR

TEACHER

Italy

2016 – To Date

Invited to offer machine learning-related classes at the International Doctorate School entitled “Scuola di Visione delle Macchine” (2016) / International Summer School on Artificial Intelligence - From Deep Learning to Data Analytics (2018, 2019).

TEACHING

National and International Universities

TEACHER

Italy/Austria

2011 – To Date

Teaching activities on bachelor, master, and doctoral computer science and industrial engineering courses offered by the University of Udine (Italy) and the Alpen-Adria University of Klagenfurt (Austria).

DOCTORATE SCHOOL ORGANIZER

International Schools

MEMBER OF THE ORGANIZING TEAM

Italy

2016-To Date

Member of the organizing team of the 2016 edition of the International Doctorate School entitled “Scuola di Visione delle Macchine” promoted by the Italian IAPR Chapter and of the International Summer School on Artificial Intelligence (AI-DLDA), 2018 to 2025 editions.

DOCTORATE THESIS REFEREE

National Referee

UNIVERSITY OF PADOVA/UDINE

Italy

2018 – To Date

External/Internal member of the evaluation committee for award of International Doctor of Philosophy Degrees.

International Referee

UNIVERSITY OF ALCALA DE HENARES

Alcala de Henares, Spain

2015 – To Date

International member of the evaluation committee for award of the International Doctor of Philosophy Degree in Electronics, department of Electronics, University of Alcala de Henares, Spain.

INTERNATIONAL JOURNAL EDITOR

Elsevier Microprocessors and Microsystems International Journal

HANDLING/ASSOCIATE EDITOR

Elsevier

2020

Handling/Associate editor for the International Journal on Microprocessors and Microsystem.

IEEE/IET CAAI Transactions on Intelligence Technology

ASSOCIATE EDITOR

IEE/IET

2016

Associate editor for the International IEEE-IET Journal CAAI Transactions on Intelligence Technology.

INTERNATIONAL RESEARCH

University of California at Riverside

VISITING RESEARCHER

Riverside, CA, USA

2014 – 2016

Junior specialist (2012–2013) and visiting assistant project scientist (2016) working on projects funded by the Office of Naval Research (ONR) and by the National Science Foundation (NSF).

Alpen-Adria University of Klagenfurt

VISITING PROFESSOR

Klagenfurt, Austria

2018/2020/2022/2024

Visiting professor for guest lectures series on Artificial Vision.

RESEARCH COLLABORATIONS

National and International Universities

RESEARCHER

Italy/Slovenia/USA/Australia/The Netherlands

2012 – To Date

Collaborations with national and international universities including: University of Firenze, University of Ljubljana, University of Boston, University of California at Riverside, Queensland University of Technology, University of Adelaide, Stanford University, Naturalis Biodiversity Center.

INTERNATIONAL CONFERENCE ORGANIZATION

International Conference on Image Analysis and Processing

PUBLICATION CHAIR

Publication chair for the 22nd International Conference on Image Analysis and Processing, held in Udine, Italy.

Udine, Italy

2023

International Workshop on Distributed Smart Cameras (ICCV2021, ECCV2022)

GENERAL CHAIR

General chair since the 1st International Workshop on Distributed Smart Cameras.

Virtual

2021-2022

International Conference on Image Analysis and Processing

SPECIAL SESSION CHAIR

Special Session Chair for the 21st International Conference on Image Analysis and Processing, held in Lecce, Italy.

Lecce, Italy

2021

International Conference on Distributed Smart Cameras

MULTIMEDIA CHAIR

Multimedia chair for the 12th International Conference on Distributed Smart Cameras, held in Eindhoven, The Netherlands.

Eindhoven, The Netherlands

2018

International Conference on Distributed Smart Cameras

PUBLICATION CHAIR

Publication chair for the 8th International Conference on Distributed Smart Cameras, held in Venice, Italy.

Venice, Italy

2014

IEEE/ACM/Springer

TECHNICAL PROGRAM COMMITTEE

Member of the technical program committee of various top-notch computer vision and machine learning international conferences and workshops (e.g., WACV/ECCV/ICCV/CVPR/AAAI/NeurIPS).

2013 - To Date

INVITED PRESENTATIONS AND SEMINARS

International Universities

SEMINARS

Invited to present current and past research activities at different International Universities, including the University of Ljubljana (Slovenia) and the University of Massachusetts at Lowell (Massachusetts, USA).

2015 - To Date

TECHNOLOGICAL TRANSFER

Eye-Tech Srl

MEMBER

Member of an (ex) spin-off of the University of Udine.

2016 - To Date

REVIEWING ACTIVITIES

IEEE/ACM/Springer

REVIEWER

Reviewer and member of the technical program committee of top-notch international conferences.

2014 - To Date

IEEE/ACM/Elsevier/IET/SPIE/Springer

REVIEWER

Reviewer for prestigious international computer vision and machine learning-related journals.

2014 - To Date

COMPETITIVE RESEARCH PROJECTS

University of Udine

COPI/WP/TECHNICAL RESPONSIBLE

PI, CoPI, WP responsible and unit member for 8 international and 5 national projects.

Udine, Italy

2010 - To Date

PUBLICATIONS

IEEE/ACM/Elsevier/Springer

AUTHOR/CO-AUTHOR

Author or co-author of more than 20 international journal papers, 50 papers published in international conferences proceedings and 3 book chapters.

2010 - To Date

IEEE/ACM

AWARDS

2014 – To Date

- Ranked among the **top reviewers for ECCV 2020** and top outstanding reviewers for CVPR 2021.
- Main author of the paper entitled “Unsupervised Hashing with Neural Trees for Image Retrieval and Person Re-Identification” by N. Martinel, G. L. Foresti and C. Micheloni, nominated for the **Best Paper Award** at the International Conference on Distributed Smart Cameras (2018)
- Main author of the paper entitled “Distributed and Unsupervised Cost-Driven Person Re-identification” by N. Martinel, G. L. Foresti and C. Micheloni, receiving the **Best Industry Related Paper Award** at the International Conference on Pattern Recognition (2016)
- Main author of the paper entitled “Sparse Matching of Random Patches for Person Re-Identification” by N. Martinel and C. Micheloni, receiving the **Best Paper Award** at the International Conference on Distributed Smart Cameras (2014)

RESEARCH QUALITY

Bibliometric Indicators

AUTHOR

2012 – To Date

- **Scopus:** h – index = 24 – citations = 2157
- **Google Scholar:** h – index = 27 – i10 – index = 51 – citations = 2933

Detailed List

Education

University of Udine

PH.D. IN INDUSTRIAL AND INFORMATION ENGINEERING

Udine, Italy

May 2014 - May 2017

- Thesis entitled "Biologically-Inspired Hierarchical Learning Architectures for Image Classification".
Supervisor: Prof. Gian Luca Foresti
Università degli Studi di Udine, May 22, 2017, pp. 155

University of Udine

PH.D. IN MULTIMEDIA COMMUNICATION

Udine, Italy

January 2011 - April 2014

- Thesis entitled "A Distributed Video Surveillance System to Track Persons in Camera Networks".
Supervisor: Dr. Christian Micheloni
Università degli Studi di Udine, April 9, 2014, pp. 163.

University of Udine

MASTER DEGREE IN MULTIMEDIA COMMUNICATION

Udine, Italy

October 2008 - November 2010

- Thesis entitled "Design and Development of a Dynamic User Interface for Video Surveillance Systems".
Supervisors: Dr. Christian Micheloni and Prof. Gian Luca Foresti
Università degli Studi di Udine, November 2010, pp. 234.
- Mark: 110/110 *cum laude*.

University of Udine

BACHELOR DEGREE IN MULTIMEDIA SCIENCE AND TECHNOLOGIES

Udine, Italy

September 2005 - September 2008

- Thesis entitled "A Portal Designed for Querying XML Documents".
Supervisors: Dr. Demis Ballis and Dr. Michele Baggi
Università degli Studi di Udine, September 2008, pp. 200.
- Mark: 104/110.

International Schools

University of Cambridge, UK / University of Catania, Italy

INTERNATIONAL COMPUTER VISION SUMMER SCHOOL ICVSS 2011 - REGISTRATION, RECOGNITION AND RECONSTRUCTION
IN IMAGES AND VIDEO

Scicli, Italy

July 2011

University of Modena and Reggio Emilia / IAPR

INTERNATIONAL SHORT SPRING SCHOOL ON SURVEILLANCE - S4

Modena, Italy

May 2011

International Experiences

University of California, Riverside, CA

ASSISTANT PROJECT SCIENTIST - SUPERVISOR: PROF. AMIT K. ROY-CHOWDHURY

Riverside, CA, USA

January 2016 - March 2016

Leading a project module on person re-identification

- Implemented a human-in-the-loop method that yields incremental learning. This is used to perform temporal model (metric learning) adaptation
- Other investigated topics: metric learning; deep learning; convex optimization; active learning; incremental learning

University of California, Riverside, CA

JUNIOR SPECIALIST - SUPERVISOR: PROF. AMIT K. ROY-CHOWDHURY

Riverside, CA, USA

September 2012 - April 2013

Jointly working on a project module on person re-identification

- Implemented an approach that tackles the re-identification objective by performing classification of pairwise camera-dependent visual feature transformations
- Other investigated topics: feature transformation; unsupervised manifold learning; machine learning; deep learning; convex/distributed optimization

Teaching Activities

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2024 - 2025

University of Udine

SMART IOT DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2024 - 2025

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2024 - 2025

Alpen-Adria University at Klagenfurt

ARTIFICIAL VISION

Teacher for the master/doctorate short course offered by the Institute of Networked and Embedded Systems

Klagenfurt, Austria

May 2024

University of Udine

SMART IOT DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2023 - 2024

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2023 - 2024

University of Udine

COMPUTER SCIENCE FOR MULTIMEDIA

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2023 - 2024

University of Udine

SMART IOT DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2022 - 2023

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2022 - 2023

University of Udine

COMPUTER SCIENCE FOR MULTIMEDIA

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2022 - 2023

Alpen-Adria University at Klagenfurt

ARTIFICIAL VISION

Teacher for the master/doctorate short course offered by the Institute of Networked and Embedded Systems

Klagenfurt, Austria

May 2022

University of Udine

SMART IOT DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2021 - 2022

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2021 - 2022

University of Udine

SMART IOT DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2020 - 2021

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2020 - 2021

International Summer School on Artificial Intelligence: From Deep Learning to Data Analytics (AI-DLDA)

GOING BEYOND CONVOLUTIONAL NEURAL NETWORKS FOR COMPUTER VISION

Invited teacher for the International Doctorate School

Udine, Italy

July 2020

Alpen-Adria University at Klagenfurt

ARTIFICIAL VISION

Teacher for the master/doctorate short course offered by the Institute of Networked and Embedded Systems

Klagenfurt, Austria

May 2020

University of Udine

SMART NETWORKED DEVICES

Teacher for the master course in Multimedia and Information Technology

Udine, Italy

2019 - 2020

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2019 - 2020

University of Udine

MULTIMEDIA DATA MANAGEMENT LAB

Teacher for the bachelor course in Multimedia and Science Technologies

Udine, Italy

2018 - 2019

International Summer School on Artificial Intelligence: From Deep Learning to Data Analytics (AI-DLDA)

TOWARDS HUMAN-LEVEL IMAGE UNDERSTANDING WITH DEEP LEARNING

Invited teacher for the International Doctorate School

Udine, Italy

June 2019

Alpen-Adria University at Klagenfurt

ARTIFICIAL VISION

Teacher for the master/doctorate course offered by the Institute of Networked and Embedded Systems

Klagenfurt, Austria

June 2018

International Summer School on Artificial Intelligence: From Deep Learning to Data Analytics (AI-DLDA)

TOWARDS HUMAN-LEVEL IMAGE UNDERSTANDING WITH DEEP LEARNING

Invited teacher for the International Doctorate School

Udine, Italy

July 2018

University of Udine

MACHINE LEARNING

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2017 - 2018

University of Udine

SMART VISION AND SENSOR NETWORKS

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2017 - 2018

Istituto Tecnico Superiore ITS - Kennedy

INTRODUCTION TO PROBABILITY

Teacher

Pordenone, Italy

2017

University of Udine

MACHINE LEARNING

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2016 - 2017

University of Udine

3D COMPUTER GRAPHICS

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2016 - 2017

University of Udine

SMART VISION AND SENSOR NETWORKS

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2016 - 2017

International Machine Vision Summer School (VISMAC)

MACHINE/DEEP LEARNING INTRODUCTION

Invited teacher for the international doctorate school

Pordenone/Grado, Italy

June 2016

University of Udine

MACHINE LEARNING

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2015 - 2016

University of Udine

3D COMPUTER GRAPHICS

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2015 - 2016

University of Udine

SMART VISION AND SENSOR NETWORKS

Commissioned teacher for the master course in Computer Science – International Degree

Udine, Italy

2015 - 2016

University of Udine

3D COMPUTER GRAPHICS

Teaching assistant for the master course in Multimedia Communication and Information Technologies

Udine, Italy

2013 - 2014

University of Udine

VIDEO PROCESSING AND TRANSMISSION

Teaching assistant for the master course in Multimedia Communication and Information Technologies

Udine, Italy

2013 - 2014

University of Udine

SECURITY IN MULTIMEDIA APPLICATIONS

Teaching assistant for the bachelor course in Multimedia Science and Technology

Udine, Italy

2013 - 2014

University of Udine

VIDEO PROCESSING AND TRANSMISSION

Teaching assistant for the master course in Multimedia Communication and Information Technologies

Udine, Italy

2011 - 2012

Working Experience

University of Udine

ASSOCIATE PROFESSOR

Udine, Italy

Dec 2021 - To Date

University of Udine

ASSISTANT PROFESSOR (TENURE TRACK)

Udine, Italy

Dec 2018 - Dec 2021

University of Udine

ASSISTANT PROFESSOR

Udine, Italy

May 2018 - Dec 2018

University of Udine

POST-DOC FELLOW - SUPERVISOR: PROF. GIAN LUCA FORESTI

Leading research projects on person re-identification and pattern recognition

- Proposed an aerial video surveillance system that exploits small-scale UAVs to perform environmental monitoring
- Designed a deep learning architecture for food recognition that exploits the vertical layer structure of food dishes
- Proposed a re-identification solution that transfer the knowledge available in the single-person re-identification domain to the group re-identification one

Udine, Italy

May 2016 - May 2018

Eye-Tech S.r.l.

ADVISORY BOARD MEMBER - MACHINE LEARNING EXPERT

Leading R&D projects on Machine Learning and Computer Vision

- Design and implementation of a prototype software for the automatic evaluation of goalkeepers
- Design and implementation of an automatic food understanding approach for embedded platforms
- Re-designed the Eye-D software to control the access of sensible areas through biometric identification
- Supervised the design and engineering of the Smart Person Identification software

Pordenone, Italy

December 2015 - May 2017

University of Udine

POST-DOC FELLOW - SUPERVISOR: PROF. GIAN LUCA FORESTI

Leading research projects on person re-identification and pattern recognition

- Proposed a real-time incremental and geo-referenced solution to build robust mosaics by means of small-scale UAVs
- Proposed a stochastic optimization-based method to temporally adapt a metric learning model with human in the loop
- Designed a neural tree-based solution that borrows from neural networks and extreme learning machines to self-discover the optimal architecture
- Implemented a person re-identification framework that optimizes the network resources and largely improves the performances of existing solutions

Udine, Italy

June 2015 - May 2016

University of Udine

Udine, Italy

POST-DOC FELLOW - SUPERVISOR: PROF. GIAN LUCA FORESTI

February 2014 - May 2015

Leading research projects on person re-identification

- Designed a re-ranking solution that discovers the visual similarities shared between similar persons and removes them to focus on the relevant re-identification details
- Designed a re-identification framework to fuse multiple experts re-identification decisions through a probabilistic scheme
- Proposed a framework that exploits visual saliency and learns multiple metrics between different image features

University of Udine

Udine, Italy

COMPUTER VISION EXPERT - SUPERVISOR: PROF. GIAN LUCA FORESTI

October 2013 - March 2014

Unit Leader for the development of a mosaicking software for aerial images acquired by low altitude Unmanned Aerial Vehicles (UAVs). National Project ADVISOR III.

- Supervised the software engineering
- Designed the incremental stitching computer vision solution that exploits GPS-coordinates to perform real time operations

InfoFactory S.r.l.

Udine, Italy

COMPUTER SYSTEMS ANALYST, DESIGNER AND DEVELOPER

June 2010 - December 2011

Design of Graphical User Interface for the SRSNet Project funded by the European Commission

- Designed the User Interface through Human-Computer Interaction evaluation procedures
- Development of the computer vision solution to track the targets across different overlapping cameras

MEDIASTUDIO S.r.l.

Pordenone, Italy

COMPUTER SYSTEMS ANALYST, DESIGNER AND DEVELOPER

April 2009 - December 2010

Design of Graphical User Interface for 3D Software | Design and development of mobile applications

- Development of mobile applications for the design of customized kitchens
- Re-design and re-engineering of the graphical user interface of the software product used to design customized kitchens

MEDIASTUDIO S.r.l.

Pordenone, Italy

COMPUTER SYSTEMS ANALYST, DESIGNER AND DEVELOPER

March 2008 - March 2009

Design of Graphical User Interface for 3D Software

- Re-design and re-engineering of the graphical user interface of the software product used to create/draw kitchens objects (i.e., a Computer Aided Design software)

FotocolorMartinelli S.n.c.

Maniago, PN, Italy

GRAPHIC DESIGNER

June 2008 - September 2008

Photo retouching | Dealer

FotocolorMartinelli S.n.c.

Maniago, PN, Italy

GRAPHIC DESIGNER

June 2007 - September 2007

Photo retouching | Dealer

National and International Collaborations

NATO

Bruxelles, Belgium

NATO ACTIVITIES IST-221 DEFENSE AGAINST ADVERSARIAL ATTACKS ON MACHINE LEARNING SYSTEMS SPACE

2025 - To Date

Determine the state-of-the-art detection and mitigation measures to counter the AI-adversarial specific threats in the military ICT infrastructures context.

NATO

Bruxelles, Belgium

NATO ACTIVITIES IST-169 ROBUSTNESS AND ACCOUNTABILITY IN MACHINE LEARNING SYSTEM

2018 - 2023

Determine the state-of-the-art in robustness and accountability for deep learning systems with complex and large models which are virtually impossible to manage by humans.

NATO

Bruxelles, Belgium

MEMBER OF THE NATO ACTIVITY IST-ET-113 UNSUPERVISED MACHINE LEARNING IN THE MILITARY DOMAIN

2019 - 2021

Define the state-of-the-art in unsupervised machine learning systems and set the stepping-stones for the future directions in the military domain.

University of Adelaide

Adelaide, Australia

PROF. GUSTAVO CARNIERO

2018 - To Date

Collaborations with Prof. Gustavo Carniero on reinforcement learning for visual tracking

Queensland University of Technology

Brisbane, Australia

DR. DAVIDE FONTANAROSA

2018 - To Date

Collaborations with Dr. Davide Fontanarosa on visual tracking in ultrasound images and robotics for surgery applications

University of Firenze

PROF. ALBERTO DEL BIMBO

Collaborations with Prof. Alberto del Bimbo on re-identification solutions proposing single-to-group transfer learning approaches

Firenze, Italy

2016 – To Date

University of Pavia

DR. GIUSEPPE LISANTI

Collaborations with Dr. Giuseppe Lisanti on re-identification solutions proposing single-to-group transfer learning approaches

Firenze, Italy

2016 – To Date

University of Ljubljana

PROF. DANJIEL SKOCAJ

Collaborations with Prof. Danjuel Skocaj on incremental learning solutions

Ljubljana, Slovenia

2015 – To Date

University of Alcalá de Henares

PROF. ALFREDO GARDEL VICENTE

Collaborations with Prof. Alfredo Gardel Vicente investigating re-ranking schemes for person re-identification

Alcalá de Henares, Spain

2014 – To Date

Indian Institute of Technology, Kharagpur

DR. ABIR DAS

Collaborations with Dr. Abir Das on person re-identification solutions

Boston, CA, USA

2012 – To Date

University of California at Riverside

PROF. AMIT K. ROY-CHOWDHURY

Collaborations with Prof. Amit K. Roy-Chowdhury investigating person re-identification solutions grounded on human-in-the-loop and feature transformation techniques. Other investigated topics include: machine learning; deep learning; convex optimization; active learning.

Riverside, CA, USA

2012 – To Date

University of Stanford

PROF. EHSAN ADELI

Collaborations with Prof. Ehsan Adeli on distributed sensors networks for content summarization and image generation.

Stanford, CA, USA

2020 – To Date

International Research Projects

EU Grant 2024 2nd Call GREEN ERA-HUB

PRINCIPAL INVESTIGATOR (PI) FOR THE RESEARCH PROJECT “FARMSCAN: FIELD ANALYSIS VIA REMOTE MONITORING AND SUSTAINABLE, DECOMPOSABLE AGRICULTURAL NODES” (FARMSCAN)

EU

2025 - 2028

Office of Naval Research

CO-PRINCIPAL INVESTIGATOR (COPI) FOR THE RESEARCH PROJECT “TARGET RE-ASSOCIATION FOR AUTONOMOUS AGENTS” (TRAAA)

USA

2020 - 2022

European Community – Marie Curie European Training Network project

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: ADVANCED HARDWARE/SOFTWARE COMPONENTS FOR INTEGRATED/EMBEDDED VISION SYSTEMS (ACHIEVE)

EU

2017 - 2021

Italian/Singapore Ministry of Defense

MEMBER/WP RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: PRESCRIPTIVE SITUATIONAL AWARENESS FOR COOPERATIVE AUTO-ORGANIZING AERIAL SENSOR NETWORKS (PRESNET)

Rome, Italy / Singapore, Singapore

2016 - 2019

National Science Foundation

MEMBER OF THE RESEARCH TEAM WORKING ON THE NATIONAL SCIENCE FOUNDATION PROJECT ENTITLED “MULTIROBOT-HUMAN COORDINATION FOR VISUAL SCENE UNDERSTANDING” – IIS-1316934

Riverside, CA, USA

2016

European Community – Interreg IV research project

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: SMART RESOURCE-AWARE MULTI-SENSOR NETWORK (SRSNET)

EU

2011 - 2012

European Community – Transnational Cooperation Programme EU-SEE/A/426/1

EU

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: INTER-REGIONAL COOPERATION FOR VALORISATION OF R&D RESULTS (INTERVALUE)

2009 - 2012

European Defense Agency – Defence R&T Joint Investment Programme on Force Protection - Project A-0444-RT-GC

EU

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: DISTRIBUTED AND ADAPTIVE MULTISENSOR FUSION ENGINE (DAFNE)

2009 - 2011

European Defense Agency – Defense R&T Joint Investment Program on Force Protection – Project A-0380-RT-GC

EU

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: MULTI-SENSOR ANTISNIPER SYSTEM (MUSAS)

2007 - 2010

National Research Projects

Italian Ministry of Education, Universities and Research / NextGeneration EU

Rome, Italy

MEMBER/WP LEADER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE FOR THE PROJECT ENTITLED: “TRACKING IN EGOVISION FOR APPLIED MEMORY”

2022 - To Date

Ergolines Lab s.r.l.

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE FOR THE PROJECT ENTITLED: “SVILUPPO, IMPLEMENTAZIONE E VALIDAZIONE DI ALGORITMI DI INTELLIGENZA ARTIFICIALE APPLICATI ALL’ANALISI DEI DATI ACQUISITI DA SENSORISTICA CON APPLICAZIONE IN LINGOTTIERA PER MACCHINA DI COLATA CONTINUA ACCIAIO”

2020 - 2022

SMACT Scpa

Padova, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE FOR THE PROJECT ENTITLED: “PIATTAFORMA IoT PER CONTROLLO ED ANALISI QUALITÀ DATA-DRIVEN IN AMBITO METALLURGICO”

2020 - 2021

Ergolines Lab s.r.l.

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE FOR THE PROJECT ENTITLED: “RICERCA E SVILUPPO DI ALGORITMI DI MACHINE LEARNING PER UN LORO UTILIZZO IN SISTEMI TOMOGRAFICI”

2020

Italian Ministry of Defense

Rome, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: A PROACTIVE COUNTER-UAV SYSTEM TO PROTECT ARMY TANKS AND PATROLS IN URBAN AREAS (PROACTIVE COUNTER-UAV)

2019 - To Date

Ergolines Lab s.r.l.

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE FOR THE PROJECT ENTITLED: “STUDI ESPLORATIVI DI SVILUPPO DI ALGORITMI DI MACHINE LEARNING PER UN LORO UTILIZZO IN SISTEMI TOMOGRAFICI”

2019 - 2020

Friuli Venezia-Giulia Region / beanTech Srl

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: SMART MACHINE BOX (SMB)

2018 - 2020

Friuli Venezia-Giulia Region / Wartsila Spa

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: ADVANCED TRACKING OF TECHNICAL PERSONNEL THROUGH AUGMENTED REALITY

2018 - 2020

Friuli Venezia-Giulia Region / Rosa Plast Srl

Trieste, Italy

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: COLLABORATIVE DESIGN WITH MIXED REALITY

2018 - 2020

Friuli Venezia-Giulia Region / Nuova Contec Srl

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: INTEGRATED VIDEO ANALYTICS SYSTEM FOR THE INSPECTION OF SEWER PIPES (SVIPCF)

Trieste, Italy

2016 - 2017

Italian Ministry of Defense

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: AUGMENTED REALITY FOR MOBILE APPLICATIONS: NAVIGATION SUPPORT FOR MILITARY PATROLS IN HIGH RISK AREAS, ADVANCED VISUALIZATION OF INTEREST POINTS IN TOURISTIC APPLICATIONS AND INTELLIGENT RECOGNITION OF EXPLOSIVE REMNANT OF WAR (RA2M)

Rome, Italy

2015 - 2017

Italian Ministry of Defense/Friuli Venezia-Giulia Region

MEMBER/TECHICAL RESPONSIBLE OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: PROACTIVE ARTIFICIAL VISION FOR ADVANCED UAV SYSTEMS FOR PROTECTION OF MILITARY UNITS, TERRITORIAL CONTROL AND ENVIRONMENTAL PREVENTION (SUPREME)

Rome/Trieste, Italy

2015 - 2017

Italian Ministry of Defense

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: SYSTEM DEPLOYMENT AND AUTOMATIC NETWORK RECONFIGURATION OF MOBILE SENSORS ON UAVS FOR THE PROTECTION OF CONVOYS, PATROLS AND MILITARY INSTALLATIONS OF (ADVISOR III)

Rome, Italy

2011 - 2013

Italian Ministry of Defense

MEMBER OF THE RESEARCH UNIT OF THE UNIVERSITY OF UDINE: AN ADVANCED MODEL FOR AUTONOMOUS RECONFIGURATION OF A NETWORK OF ACTIVE SENSORS FOR PROTECTION OF MILITARY CAMPS IN ABROAD MISSIONS (ADVISOR II)

Rome, Italy

2008 - 2011

Journal Editor

Elsevier Microprocessors and Microsystems International Journal

HANDLING/ASSOCIATE EDITOR

Handling/Associate editor for the International Journal on Microprocessors and Microsystem.

Elsevier

2020 - To Date

CAAI Transactions on Intelligence Technology

ASSOCIATE EDITOR

Associate editor for the International IEEE-IET Journal CAAI Transactions on Intelligence Technology.

IEEE / IET / Wiley & Sons

2016 - To Date

Reviewing Activities

IEEE

- JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING
- TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE
- TRANSACTIONS ON IMAGE PROCESSING
- TRANSACTIONS ON CYBERNETICS
- TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY
- TRANSACTIONS ON BIOMETRICS, BEHAVIOR, AND IDENTITY SCIENCE
- TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY
- TRANSACTIONS ON INTELLIGENT SYSTEMS AND TECHNOLOGY
- TRANSACTIONS ON MULTIMEDIA
- SIGNAL PROCESSING LETTERS
- ROBOTICS AND AUTOMATION LETTERS
- IEEE ACCESS

Elsevier

- IMAGE AND VISION COMPUTING
- PATTERN RECOGNITION
- COMPUTER VISION AND IMAGE UNDERSTANDING
- PATTERN RECOGNITION LETTERS
- SIGNAL PROCESSING: IMAGE COMMUNICATION
- JOURNAL OF VISUAL COMMUNICATION AND IMAGE REPRESENTATION REGISTRATION
- INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS
- KNOWLEDGE-BASED SYSTEMS
- 3D RESEARCH
- NEUROCOMPUTING

ACM

- TRANSACTION ON SENSOR NETWORKS

Springer

- THE VISUAL COMPUTER
- INTERNATIONAL JOURNAL OF COMPUTER VISION
- NEURAL COMPUTING AND APPLICATIONS

IET

- COMPUTER VISION
- IMAGE PROCESSING

SPIE

- JOURNAL OF ELECTRONICAL IMAGING
- OPTICAL ENGINEERING

International Conference/Workshop Organization

GENERAL CHAIR

International Wokrhsop on Distributed Smart Cameras (IWDSC)

Virtual
September 2022

International Wokrhsop on Distributed Smart Cameras (IWDSC)

Virtual
October 2021

SPECIAL SESSION CHAIR

International Conference on Image Analysis and Processing (ICIAP)

Lecce, Italy
May 2022

MULTIMEDIA CHAIR

ACM International Conference on Distributed Smart Cameras (ICDSC)

Eindhoven, The Netherlands
September 2018

ACM International Conference on Distributed Smart Cameras (ICDSC)

Stanford, CA, USA
September 2017

PUBLICATION CHAIR

International Conference on Image Analysis and Processing (ICIAP)

Udine, Italy
September 2023

IEEE/ACM International Conference on Distributed Smart Cameras (ICDSC)

Venice, Italy
November 2014

PANEL SESSION CHAIR

IEEE/ACM International Conference on Distributed Smart Cameras (ICDSC)

Venice, Italy
November 2014

LOCAL CHAIR

International Machine Vision Summer School (VISMAC)

Pordenone/Grado, Italy
June 2016

International Conference for the Italian Chapter on Pattern Recognition (GIRPR)

Grado (UD), Italy
June 2016

AREA CHAIR

International Conference on Computer Vision and Pattern Recognition (CVPR)

Denver, USA
June 2026

International Conference on Computer Vision and Pattern Recognition (CVPR)

Nashville, USA
June 2025

European Conference on Computer Vision (ECCV)

Milan, Italy
October 2024

Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)

Gandhinagar, India
December 2022

International Conference on Pattern Recognition (ICPR)

Montreal, Canada
August 2022

International Conference on Pattern Recognition (ICPR)

Milano, Italy
January 2021

TECHNICAL PROGRAM COMMITTEE

IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)

Seattle, WA, USA
June 2024

IAPR International Conference on Advanced Robotics (ICAR)

Abu Dhabi, UAE
December 2023

Conference on Neural Information Processing Systems (NeurIPS)

New Orleans, Louisiana, USA
December 2023

IEEE International Conference on Computer Vision (ICCV)	<i>Paris, France</i> October 2023
IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)	<i>Vancouver, Canada</i> June 2023
International Conference on Learning Representations (ICLR)	<i>Hybrid, Kigali, Rwanda</i> May 2023
European Conference on Computer Vision (ECCV)	<i>Tel Aviv, Israel</i> October 2022
IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)	<i>New Orleans, Louisiana, USA</i> June 2022
IEEE International Conference on Computer Vision (ICCV)	<i>Virtual</i> October 2021
IEEE International Conference on Computer Vision and Pattern Recognition (CVPR) - Outstanding Reviewer	<i>Virtual</i> June 2021
AAAI Conference on Artificial Intelligence (AAAI)	<i>Virtual</i> February 2021
IEEE International Conference on Winter Applications of Computer Vision (WACV)	<i>Virtual</i> January 2021
Asian Conference on Computer Vision (ACCV)	<i>Kyoto, Japan</i> November-December 2020
European Conference on Computer Vision (ECCV) - Outstanding Reviewer	<i>Edinburgh, Scotland</i> August 2020
IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)	<i>Seattle WA, USA</i> June 2020
AAAI Conference on Artificial Intelligence (AAAI)	<i>New York NY, USA</i> February 2020
IEEE International Conference on Robotics and Automation (ICRA)	<i>Paris, France</i> May-June 2020
IEEE International Conference on Computer Vision (ICCV)	<i>Seoul, South Korea</i> October-November 2019
IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)	<i>Long Beach CA, USA</i> June 2019

IEEE International Conference on Distributed Smart Cameras (ICDSC)	<i>Trento, Italy</i> September 2019
IEEE International Conference on Robotics and Automation (ICRA)	<i>Montreal, Canada</i> May 2019
IEEE International Conference on Winter Applications of Computer Vision (WACV)	<i>Waikoloa Village, HI, USA</i> January 2019
IEEE International Conference on Distributed Smart Cameras (ICDSC)	<i>Eindhoven, The Netherlands</i> September 2018
International Conference on Pattern Recognition (ICPR)	<i>Beijing, China</i> August 2018
IEEE International Conference on Winter Applications of Computer Vision (WACV)	<i>Lake Tahoe, CA, USA</i> March 2018
International Workshop on on Cross-domain Human Identification (CHI)	<i>Venice, Italy</i> October 2017
International Workshop on Multimedia Assisted Dietary Management (MADIMA)	<i>Catania, Italy</i> December 2017
IEEE International Conference on Image Processing Theory, Tools and Applications (IPTA)	<i>Montreal, Canada</i> December 2017
IEEE International Conference on Computer Vision (ICCV)	<i>Venice, Italy</i> October 2017
International Workshop on Assistive Computer Vision and Robotics (ACVR)	<i>Venice, Italy</i> October 2017
IEEE International Conference on Distributed Smart Cameras (ICDSC)	<i>Stanford, CA, USA</i> September 2017
International Symposium on Image and Signal Processing and Analysis (ISPA)	<i>Ljubljana, Slovenia</i> September 2017
IEEE International Conference on Advanced Video and Signal based Surveillance (AVSS)	<i>Lecce, Italy</i> September 2017
IEEE International Conference on Winter Applications of Computer Vision (WACV)	<i>Santa Rosa, CA, USA</i> March 2017
International Conference on Pattern Recognition (ICPR)	<i>Cancun, Mexico</i> December 2016

IEEE International Workshop on Assisted Computer Vision for Robotics (ACVR) - in Conjunction with ECCV

Amsterdam, The Netherlands

October 2016

ACM International Conference on Distributed Smart Cameras (ICDSC)

Paris, France

September 2016

ACM International Conference on Distributed Smart Cameras (ICDSC)

Seville, Spain

September 2015

IEEE/ACM International Conference on Distributed Smart Cameras (ICDSC)

Venice, Italy

November 2014

Skills

Machine Learning

- NEURAL NETWORKS/DEEP LEARNING, METRIC LEARNING, NEURAL TREES, EXTREME LEARNING MACHINES, SUPPORT VECTOR MACHINES, DICTIONARY LEARNING, INCREMENTAL LEARNING, DIMENSIONALITY REDUCTION, OPTIMIZATION
- ENVIRONMENTS/TOOLS: PYTORCH, PYTHON, C/C++, ANACONDA, PYCHARM, MATLAB

Computer Vision

- VISUAL UNDERSTANDING/RECOGNITION/DETECTION/TRACKING, FEATURE LEARNING, FEATURE EXTRACTION, FEATURE MATCHING, TEMPORAL ADAPTATIONS
- ENVIRONMENTS/TOOLS: PYTORCH, PYTHON, C#, C/C++, SPYDER, PYCHARM, MATLAB, VISUAL STUDIO

Unmanned vehicles

- UGV, UAV, AUTONOMOUS NAVIGATION, VISUAL ANALYTICS
- ENVIRONMENTS/TOOLS: ROS, PYTHON, C#, C/C++, SPYDER, PYCHARM, VISUAL STUDIO, ARDUINO, RASPBERRYPI

Programming

- LANGUAGES: PYTHON, C/C++, C#, OBJECTIVE-C, LUA, MATLAB, SQL
- ENVIRONMENTS/TOOLS: ANACONDA, SPYDER, PYCHARM, VISUAL STUDIO, XCODE, ZEROBRANE STUDIO, SQL SERVER, MYSQL

Web

HTML5, AJAX, PHP, JAVASCRIPT, CSS, SQL

Languages

- ENGLISH: HIGHLY PROFICIENT IN SPOKEN AND WRITTEN
- ITALIAN: NATIVE

Scientific Associations / International Organizations

NATO

MEMBER OF THE NATO ACTIVITY IST-169 ROBUSTNESS AND ACCOUNTABILITY IN MACHINE LEARNING SYSTEM

Bruxelles, Belgium

August 2018 - To Date

NATO

MEMBER OF THE NATO ACTIVITY IST-ET-113 UNSUPERVISED MACHINE LEARNING IN THE MILITARY DOMAIN

Bruxelles, Belgium

December 2019 - To Date

IAPR

DEPUTY CHIEF FOR THE NEURAL NETWORKS AND DEEP LEARNING TECHNICAL COMMITTEE – INTERNATIONAL ASSOCIATION FOR PATTERN RECOGNITION, ITALIAN CHAPTER

2017 - To Date

IEEE

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS MEMBER

2011 - To Date

IAPR

MEMBER OF THE GRUPPO ITALIANO RICERCATORI IN PATTERN RECOGNITION (GIRPR) - AFFILIATED WITH INTERNATIONAL ASSOCIATION OF PATTERN RECOGNITION

2011 - To Date

Awards and Small Grants

NATO IST-190 Research Symposium on AI, ML and BD for Hybrid Military Operations (AI4HMO)

BEST PAPER AWARD

October 2021

Award received for the paper entitled "Robustness of Artificial Intelligence for Hybrid Warfare"

European Conference on Computer Vision

OUTSTANDING REVIEWER

September 2020

Award received for having provided more than 7 excellent reviews

International Conference on Distributed Smart Cameras

Eindhoven, The Netherlands

BEST PAPER AWARD NOMINATION

September 2018

Nomination received for the work entitled "Unsupervised Hashing with Neural Trees for Image Retrieval and Person Re-Identification" by N.

Martinel, G. L. Foresti and C. Micheloni

NVIDIA Corporation

Santa Clara, CA, USA

RESEARCH DONATION

June 2018

Donation of an NVIDIA GPU for a research project on Unsupervised Learning with Neural Trees

International Conference on Pattern Recognition

Cancun, Mexico

BEST INDUSTRY RELATED PAPER AWARD (BIRPA)

December 2016

Award received for the work entitled "Distributed and Unsupervised Cost-Driven Person Re-identification" by N. Martinel, G. L. Foresti and C.

Micheloni

International Conference on Pattern Recognition

Cancun, Mexico

BEST STUDENT PAPER AWARD CANDIDATE

December 2016

"Distributed and Unsupervised Cost-Driven Person Re-identification" by N. Martinel, G. L. Foresti and C. Micheloni

NVIDIA Corporation

Santa Clara, CA, USA

RESEARCH DONATION

May 2015

Donation of an NVIDIA GPU for a research project on Neural Trees, Convolutional Neural Networks and Extreme Learning Machines

International Conference on Distributed Smart Cameras

Venice, Italy

BEST PAPER AWARD

September 2014

Award received for the work entitled "Sparse Matching of Random Patches for Person Re-Identification" by N. Martinel and C. Micheloni

Invited Seminars and Talks

BRIC-INAIL

Rome, Italy

COMPUTER VISION FOR MOTION ANALYSIS

March 2025

Deep Learning Meetup

Ljubljana, Slovenia

DEEP MACHINE VISION

April 2019

Deep Learning Meetup

Ljubljana, Slovenia

NUTS AND BOLTS OF MOBILE FOOD UNDERSTANDING

April 2019

Office of Naval Research/National Research Laboratory

Washington, D.C., USA

A JOURNEY INTO PERSON RE-IDENTIFICATION

February 2019

University of Milano-Bicocca BIOLOGICALLY-INSPIRED HIERARCHICAL LEARNING ARCHITECTURES FOR FOOD UNDERSTANDING	<i>Milano, Italy</i> <i>February 2018</i>
University of Milano-Bicocca FILL THE GAP FOR NON-CONTINUOUS LONG-TERM TRACKING	<i>Milano, Italy</i> <i>February 2018</i>
University of Udine DEEP LEARNING: STATE-OF-THE-ART AND PERSPECTIVES	<i>Udine, Italy</i> <i>September 2017</i>
University of Massachusetts at Lowell FILL THE GAP FOR NON-CONTINUOUS LONG-TERM TRACKING	<i>Lowell, MA, USA</i> <i>April 2016</i>
University of Ljubljana ADVANCES IN PERSON RE-IDENTIFICATION	<i>Ljubljana, Slovenia</i> <i>September 2015</i>

Post-Docs Supervision

Vitalii Emelianov NUTRITION PREDICTIONS FROM FOOD IMAGES WITH FOUNDATION MODELS	<i>University of Udine, Italy</i> <i>2024–To Date</i>
Rita Pucci CAPSULE-BASED GENERATIVE MODELS FOR IMAGE COLORIZATION AND UNDERWATER ENHANCEMENT	<i>University of Udine, Italy</i> <i>2019-2022</i>

PhD Student Supervision

Talha Hanif Butt EFFICIENT SPATIAL AND TEMPORAL VIDEO SUPER-RESOLUTION	<i>University of Udine, Italy</i> <i>Expected Graduation 2028</i>
Sameer Malik NOVEL PARADIGMS FOR PARAMETER EFFICIENT FINETUNING IN FEDERATED LEARNING	<i>University of Udine, Italy</i> <i>Expected Graduation 2028</i>
Irene Gennaralli SKELETON-BASED MOTION ANALYSIS FOR MEDICAL REHABILITATION	<i>University of Udine, Italy</i> <i>Expected Graduation 2028</i>
Denis Tavaris COMPUTER VISION FOR ENVIRONMENTAL RIVER ANALYSIS THROUGH UAVS Co-Supervised with Prof. Gian Luca Foresti	<i>University of Udine, Italy</i> <i>Expected Graduation 2027</i>
Nasar Iqbal VISUAL BASED ANOMALY DETECTION	<i>University of Udine, Italy</i> <i>Expected Graduation 2027</i>
Mubashara Rehman ARTIFICIAL INTELLIGENCE AND DATA ANALYSIS FOR PERSONALIZED ONCOLOGY Co-Supervised with Prof. Christian Micheloni	<i>University of Udine, Italy</i> <i>Expected Graduation 2026</i>
Asif Hussain Khan DEEP VIDEO SUPER-RESOLUTION	<i>University of Udine, Italy</i> <i>Graduated July 2025</i>

Publications

INTERNATIONAL JOURNALS

Lightweight Prompt Learning Implicit Degradation Estimation Network for Blind Super Resolution

KHAN, ASIF HUSSAIN AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI
IEEE Transactions on Image Processing

In Press

Lightweight Implicit Blur Kernel Estimation Network for Blind Image Super-Resolution

KHAN, ASIF HUSSAIN AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI
Information

14 (5) 2023

Deep convolutional feature details for better knee disorder diagnoses in magnetic resonance images

DUNNHOFER, MATTEO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN
Computerized Medical Imaging and Graphics

(102) 2022

Consistent attentive dual branch network for person re-identification

MUNIR, ASAD AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN
Multimedia Tools and Applications

17 (81) 2022

Weakly-Supervised Domain Adaptation of Deep Regression Trackers via Reinforced Knowledge Distillation

DUNNHOFER, MATTEO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN
IEEE Robotics and Automation Letters

2021

Lord of the Rings: Hanoi Pooling and Self-Knowledge Distillation for Fast and Accurate Vehicle Re-Identification

MARTINEL, NIKI AND DUNNHOFER, MATTEO AND PUCCI, RITA AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN
IEEE Transactions on Industrial Informatics

AUGUST 2021

An Intelligent Auto-Organizing Aerial Robotic Sensor Network System for Urban Surveillance

LEONG, WAI LUN AND MARTINEL, NIKI AND HUANG, SUNAN AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND TEO, RODNEY SWEE HUAT
Journal of Intelligent and Robotic Systems

2 (102) 2021

Deep interactive encoding with capsule networks for image classification

PUCCI, RITA AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI
Multimedia Tools and Applications

43-44 (79) 2020

Siam-U-Net: encoder-decoder siamese network for knee cartilage tracking in ultrasound images

DUNNHOFER, MATTEO AND ANTICO, MARIA AND SASAZAWA, FUMIO AND TAKEDA, YU AND CAMPS, SASKIA AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND CARNEIRO, GUSTAVO AND FONTANAROSA, DAVIDE
Medical Image Analysis

(60) 2020

A UAV Video Dataset for Mosaicking and Change Detection From Low-Altitude Flights

AVOLA, DANILO AND CINQUE, LUIGI AND FORESTI, GIAN LUCA AND MARTINEL, NIKI AND PANNONE, DANIELE AND PICIARELLI, CLAUDIO
IEEE Transactions on Systems, Man, and Cybernetics: Systems

6 (50) 2020

Deep Pyramidal Pooling With Attention for Person Re-Identification

MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN
IEEE Transactions on Image Processing

(29) 2020

From person to group re-identification via unsupervised transfer of sparse features

LISANTI, GIUSEPPE AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND DEL BIMBO, ALBERTO AND LUCA FORESTI, GIAN
Image and Vision Computing

(83-84) 2019

Distributed person re-identification through network-wise rank fusion consensus

MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN
Pattern Recognition Letters

(124) 2019

Coverage Control of Multiple Unmanned Aerial Vehicles: A Short Review

HUANG, SUNAN AND TEO, RODNEY AND LEONG, WILLIAM AND MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN
Unmanned Systems

2 (6) 2018

<i>Accelerated low-rank sparse metric learning for person re-identification</i>	
MARTINEL, NIKI	(112) 2018
Pattern Recognition Letters	
<i>An Ensemble Feature Method for Food Classification</i>	
MARTINEL, NIKI AND PICIARELLI, CLAUDIO AND MICHELONI, CHRISTIAN	1 (26) 2017
Machine Graphics and Vision	
<i>Discriminant Context Information Analysis for Post-Ranking Person Re-Identification</i>	
GARCIA, JORGE AND MARTINEL, NIKI AND GARDEL, ALFREDO AND BRAVO, IGNACIO AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN	4 (26) 2017
IEEE Transactions on Image Processing	
<i>Person Reidentification in a Distributed Camera Network Framework</i>	
MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN	11 (47) 2017
IEEE Transactions on Cybernetics	
<i>A supervised extreme learning committee for food recognition</i>	
MARTINEL, NIKI AND PICIARELLI, CLAUDIO AND MICHELONI, CHRISTIAN	(148) 2016
Computer Vision and Image Understanding	
<i>Modeling feature distances by orientation driven classifiers for person re-identification</i>	
GARCÍA, JORGE AND MARTINEL, NIKI AND GARDEL, ALFREDO AND BRAVO, IGNACIO AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN	(38) 2016
Journal of Visual Communication and Image Representation	
<i>A Pool of Multiple Person Re-Identification Experts</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA	(71) 2016
Pattern Recognition Letters	
<i>On a Distributed Video Surveillance System to Track Persons in Camera Networks</i>	
MARTINEL, NIKI	3 (14) 2015
ELCVIA Electronic Letters on Computer Vision and Image Analysis	
<i>Kernelized Saliency-Based Person Re-Identification Through Multiple Metric Learning</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA	12 (24) 2015
IEEE Transactions on Image Processing	
<i>The Evolution of Neural Learning Systems: A Novel Architecture Combining the Strengths of NTs, CNNs, and ELMs</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA	3 (1) 2015
IEEE Systems, Man, and Cybernetics Magazine	
<i>Classification of Local Eigen-Dissimilarities for Person Re-Identification</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN	4 (22) 2015
IEEE Signal Processing Letters	
<i>Re-Identification in the Function Space of Feature Warps</i>	
MARTINEL, NIKI AND DAS, ABIR AND MICHELONI, CHRISTIAN AND ROY-CHOWDHURY, AMIT K.	8 (37) 2015
IEEE Transactions on Pattern Analysis and Machine Intelligence	
<i>Person re-identification by modelling principal component analysis coefficients of image dissimilarities</i>	
MARTINEL, N. AND MICHELONI, C.	14 (50) 2014
Electronics Letters	
<i>Camera Selection for Adaptive Human-Computer Interface</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PICIARELLI, CLAUDIO AND FORESTI, GIAN LUCA	5 (44) 2014
IEEE Transactions on Systems, Man, and Cybernetics: Systems	
<i>Robust Painting Recognition and Registration for Mobile Augmented Reality</i>	
MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA	11 (20) 2013
IEEE Signal Processing Letters	

Multi-signature based person re-identification

MARTINEL, N. AND FORESTI, G.L.
Electronics Letters

13 (48) 2012

INTERNATIONAL CONFERENCES

CE-VAE: Capsule Enhanced Variational AutoEncoder for Underwater Image Enhancement

PUCCI, RITA AND MARTINEL, NIKI

February, 2025

Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision

Ntire 2024 challenge on image super-resolution (x4): Methods and results

CHEN, ZHENG AND WU, ZONGWEI AND ZAMFIR, EDUARD AND ZHANG, KAI AND ZHANG, YULUN AND TIMOFTE, RADU AND YANG, XIAOKANG AND YU, HONGYUAN AND WAN, CHENG AND HONG, YUXIN AND [...] MARTINEL, NIKI

June, 2024

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition

IDENet: Implicit Degradation Estimation Network for Efficient Blind Super Resolution

KHAN, ASIF HUSSAIN AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI

June, 2024

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition

Tracking Skiers from the Top to the Bottom

DUNNHOFER, MATTEO AND SORDI, LUCA AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN

Jan, 2024

Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision

LBKNet: Lightweight Blur Kernel Estimation Network for Blind Image Super-Resolution

KHAN, ASIF HUSSAIN AND UMER, RAO MUHAMMAD AND DUNNHOFER, MATTEO AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI

Sept, 2023

International Conference on Image Analysis and Processing

The tenth visual object tracking vot2022 challenge results

KRISTAN, MATEJ AND LEONARDIS, ALEŠ AND MATAS, JIŘÍ AND FELSBERG, MICHAEL AND PFLUGFELDER, ROMAN AND KÄMÄRÄINEN, JONI-KRISTIAN AND CHANG, HYUNG JIN AND DANELLJAN, MARTIN AND ZAJC, LUKA ČEHOVIN AND LUKEŽIČ, ALAN

Oct, 2022

European Conference on Computer Vision Workshops (ECCVW)

Spatio-Temporal Attention for Cloth-Changing ReID in Videos

BANSAL, VAIBHAV AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI

Oct, 2022

IEEE/CVF European Conference on Computer Vision Workshops (ECCVW)

Cloth-Changing Person Re-identification with Self-Attention

BANSAL, VAIBHAV AND FORESTI, GIAN LUCA AND MARTINEL, NIKI

Jan, 2022

2022 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)

Pro-CCaps: Progressively Teaching Colourisation to Capsules

PUCCI, RITA AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI

Jan, 2022

2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

CVGAN: Image Generation with Capsule Vector-VAE

PUCCI, RITA AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI

Sept, 2022

International Conference on Image Analysis and Processing

Improving MRI-based Knee Disorder Diagnosis with Pyramidal Feature Details

DUNNHOFER, MATTEO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN

Sept, 2021

Medical Imaging with Deep Learning

Collaborative Image and Object Level Features for Image Colourisation

PUCCI, RITA AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI

Jun, 2021

2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)

Self-Attention Agreement Among Capsules

PUCCI, RITA AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI

Oct, 2021

2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)

Where Did I See It? Object Instance Re-Identification with Attention BANSAL, VAIBHAV AND FORESTI, GIAN LUCA AND MARTINEL, NIKI 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)	Oct, 2021
Oriented Splits Network to Distill Background for Vehicle Re-Identification MUNIR, ASAD AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN 2021 17th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)	Nov, 2021
An Exploration of Target-Conditioned Segmentation Methods for Visual Object Trackers DUNNHOFER, MATTEO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN European Conference on Computer Vision Workshops	Aug, 2020
Multi Branch Siamese Network for Person Re-Identification MUNIR, ASAD AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN International Conference on Image Processing	Oct, 2020
Is It a Plausible Colour ? UCapsNet for Image Colourisation PUCCI, RITA AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI Neural Information Processing Systems Workshops	Dec, 2020
Self and Channel Attention Network for Person Re-Identification MUNIR, ASAD AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN International Conference on Pattern Recognition	2020
Fixed simplex coordinates for angular margin loss in CapsNet PUCCI, RITA AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA AND MARTINEL, NIKI International Conference on Pattern Recognition	2020
Tracking-by-Trackers with a Distilled and Reinforced Model DUNNHOFER, MATTEO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN Asian conference on Computer Vision	Jul, 2020
Integrated Perception and Tactical Behaviours in an Auto-Organizing Aerial Sensor Network LEONG, W L AND MARTINEL, N AND HUANG, S AND MICHELONI, C AND FORESTI, G L AND TEO, R International Conference on Unmanned Aircraft Systems	2020
The Seventh Visual Object Tracking VOT2019 Challenge Results KRISTAN, M. AND MATAS, J. AND LEONARDIS, A AND FELSBERG, M. AND PFLUGFELDER, R. AND KAMARAINEN, J-K. AND CEHOVIN, L. AND DRBOHLAV, O. AND LUKEZIC, A. AND BERG, A. AND ELDESOKEY, A. International Conference on Computer Vision Workshops	2019
Visual Tracking by means of Deep Reinforcement Learning and an Expert Demonstrator DUNNHOFER, MATTEO AND MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN International Conference on Computer Vision Workshops	2019
Video-Based Convolutional Attention for Person Re-Identification ZAMPROGNO, MARCO AND PASSON, MARCO AND MARTINEL, NIKI AND SERRA, GIUSEPPE AND MICHELONI, CHRISTIAN AND TASSO, CARLO AND FORESTI, GIAN LUCA International Conference on Image Analysis and Processing	2019
An Exploration of the Interaction Between capsules with ResNetCaps models PUCCI, RITA AND MICHELONI, CHRISTIAN AND ROBERTO, VITO AND FORESTI, GIAN LUCA AND MARTINEL, NIKI International Conference on Distributed Smart Cameras	2019
An Efficient Artificial Intelligence Framework for UAV Systems TOGOTOOTOKH, ENKHTOGTOKH AND HUANG, SUNAN AND LEONG, WAI LUN AND TEO, RODNEY AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI International Conference on Ubi-Media Computing	2019
Aggregating Deep Pyramidal Representations for Person Re-Identification MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN International Conference on Computer Vision and Pattern Recognition	2019

Unsupervised Hashing with Neural Trees for Image Retrieval and Person Re-Identification

MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN

2018

International Conference on Distributed Smart Cameras

Unsupervised Smoke Detection in Normally Smoking Environments

CHINI, MATTEO AND MARTINEL, NIKI AND DUNNHOFER, MATTEO AND CESCHIA, CARLO AND MICHELONI, CHRISTIAN

2018

International Conference on Distributed Smart Cameras

Wide-Slice Residual Networks for Food Recognition

MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN

2018

Winter Conference on Applications of Computer Vision

Person Re-Identification via Unsupervised Transfer of Learned Visual Representations

MARTINEL, NIKI AND DUNNHOFER, MATTEO AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN

2017

International Conference on Distributed Smart Cameras

Temporally Smoothed Anomaly Detection in Continuous Fluids

CHINI, MATTEO AND MARTINEL, NIKI AND SPAGNUL, STEFANO AND MICHELONI, CHRISTIAN

2017

International Conference on Distributed Smart Cameras

Real-Time Incremental and Geo-Referenced Mosaicking by Small-Scale UAVs

AVOLA, DANILO AND FORESTI, GIAN LUCA AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PANNONE, DANIELE AND

2017

PICIARELLI, CLAUDIO

International Conference on Image Analysis and Processing

Aerial Video Surveillance System for Small-Scale UAV Environment Monitoring

AVOLA, DANILO AND FORESTI, GIAN LUCA AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PANNONE, DANIELE AND

2017

PICIARELLI, CLAUDIO

Advanced Video and Signal-Based Surveillance

Group Re-Identification via Unsupervised Transfer of Sparse Features Encoding

LISANTI, GIUSEPPE AND MARTINEL, NIKI AND DEL BIMBO, ALBERTO AND FORESTI, GIAN LUCA

2017

International Conference on Computer Vision

Mobile Food Recognition with an Extreme Deep Tree

MARTINEL, NIKI AND PICIARELLI, CLAUDIO AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN

2016

International conference on Distributed Smart Cameras

Distributed and Unsupervised Cost-Driven Person Re-identification

MARTINEL, NIKI AND FORESTI, GIAN LUCA AND MICHELONI, CHRISTIAN

2016

International Conference on Pattern Recognition

Temporal Model Adaptation for Person Re-Identification

MARTINEL, NIKI AND DAS, ABIR AND MICHELONI, CHRISTIAN AND ROY-CHOWDHURY, AMIT K.

2016

European Conference on Computer Vision

Selection of temporal features for event detection in smart security

MARTINEL, N. AND AVOLA, D. AND PICIARELLI, C. AND MICHELONI, C. AND VERNIER, M. AND CINQUE, L. AND FORESTI, G.L.

2015

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On filter banks of texture features for mobile food classification

MARTINEL, N. AND PICIARELLI, C. AND MICHELONI, C. AND FORESTI, G.L.

2015

ACM International Conference Proceeding Series

Person re-identification ranking optimisation by discriminant context information analysis

GARCIA, J. AND MARTINEL, N. AND MICHELONI, C. AND GARDEL, A.

2015

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A Structured Committee for Food Recognition

MARTINEL, NIKI AND PICIARELLI, CLAUDIO AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA

2015

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Person Re-Identification Ranking Optimisation by Discriminant Context Information Analysis

GARCIA, JORGE AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND GARDEL, ALFREDO
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On Filter Banks of Texture Features for Mobile Food Classification

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Selection of Temporal Features for Event Detection in Smart Security

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Sparse Matching of Random Patches for Person Re-Identification

MARTINEL, NIKI AND MICHELONI, CHRISTIAN
International Conference on Distributed Smart Cameras

2014

Mobile Re-Identification Based on Local Features Analysis

AKHTAR, ZAHID AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA
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2014

Saliency Weighted Features for Person Re-identification

MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA
European Conference Computer Vision Workshops and Demonstrations

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Person Orientation and Feature Distances Boost Re-identification

GARCIA, JORGE AND MARTINEL, NIKI AND FORESTI, GIAN LUCA AND GARDEL, ALFREDO AND MICHELONI, CHRISTIAN
International Conference on Pattern Recognition

aug, 2014

Image processing supports HCI in museum application

MARTINEL, N. AND VERNIER, M. AND FORESTI, G.L. AND LAMEDICA, E.
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Remote feature learning for mobile re-identification

VERNIER, MARCO AND MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND FORESTI, GIAN LUCA
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Learning pairwise feature dissimilarities for person re-identification

MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PICIARELLI, CLAUDIO
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oct, 2013

Outdoor environment monitoring with unmanned aerial vehicles

PICIARELLI, CLAUDIO AND MICHELONI, CHRISTIAN AND MARTINEL, NIKI AND VERNIER, MARCO AND FORESTI, GIAN LUCA
International Conference on Image Analysis and Processing

2013

Image Processing Supports HCI in Museum Application

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feb, 2013

Distributed signature fusion for person re-identification

MARTINEL, N. AND MICHELONI, C. AND PICIARELLI, C.
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Re-identify people in wide area camera network

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jun, 2012

A Multi-Modal Self-Reconfigurable Surveillance System

MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PICIARELLI, CLAUDIO AND FORESTI, GIAN LUCA
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may, 2012

Pre-Emptive camera activation for Video Surveillance HCI

MARTINEL, NIKI AND MICHELONI, CHRISTIAN AND PICIARELLI, CLAUDIO
International Conference on Image Analysis and Processing

sep, 2011

BOOK CHAPTERS

Low-Level Feature Detectors and Descriptors for Smart Image and Video Analysis: A Comparative Study

AVOLA, D., CINQUE, L., FORESTI, G. L., MARTINEL, N., PANNONE, D., AND PICIARELLI, C.
Bridging the Semantic Gap in Image and Video Analysis – H. Kwaśnicka and L. C. Jain (Eds.)

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Distributed mobile computer vision: Advances, challenges and applications

MARTINEL, N., PRATI, A., AND MICHELONI, C
Distributed Embedded Smart Cameras: Architectures, Design and Applications – C. Bobda and S. Velipasalar (Eds.)

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2014

Advanced artificial vision and mobile devices for new applications in learning, entertainment and cultural heritage domains

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Interdisciplinary Mechatronics: Engineering Science and Research Development – M. K. Habib and J. P. Davim (Eds.)

Wiley-ISTE

2013

I states, to be informed, in accordance with and for purposes of the Italian Art. 13 of Legislative Decree 196/2003, that the collected personal data will be treated, including by computer, in proceedings to which this declaration is made. Udine, October 7, 2026